

Sumanth Kumar Lingabathini

785-813-4132 | sumanthkumar0527@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Data Analytics Scientist with 5+ years of experience applying exploratory data analysis, statistical modeling, machine learning, deep learning, and AI-enabled analytics to research, manufacturing, operational, and business forecasting problems. Proficient in Python, SQL, R, PySpark, TensorFlow, PyTorch, scikit-learn, Databricks, and large-scale data preparation. Experienced collaborating with technical subject matter experts to test hypotheses, develop analytical models, identify process patterns, and communicate findings. Knowledgeable in computer vision, CNNs, RNNs, transformers, foundation models, generative AI, LLMs, RAG, prompt engineering, model evaluation, and MLOps practices.

TECHNICAL SKILLS

Analytics & Programming: Python, SQL, R, PySpark, NumPy, Pandas, SciPy, Statistical Analysis, EDA, Hypothesis Testing, Forecasting
Machine Learning: scikit-learn, TensorFlow, PyTorch, Regression, Classification, Clustering, Feature Engineering, Anomaly Detection, Model Validation
Deep Learning & AI: CNNs, RNNs, Transformers, Computer Vision, Self-Supervised Learning Concepts, Foundation Models, Generative AI, LLMs, RAG, Prompt Engineering
Platforms & Delivery: Databricks, Spark, MLflow, Power BI, Tableau, Git, Data Mining, Large-Scale Datasets, MLOps, Model Monitoring, Technical Documentation

PROFESSIONAL EXPERIENCE

University of Kansas

Lawrence, KS

Research Data Scientist

Jan 2024 – Present

- Prepared and analyzed large research and technical datasets using Python, SQL, R, and statistical methods, improving data readiness and analytical turnaround by 35–40%.
- Performed exploratory data analysis, feature engineering, hypothesis testing, predictive modeling, and anomaly detection to identify meaningful patterns and support project-level decisions.
- Developed machine learning workflows with scikit-learn and TensorFlow/PyTorch concepts for classification, regression, forecasting, and model evaluation, improving analytical reproducibility by 30–35%.
- Collaborated with faculty and technical stakeholders to frame research questions, validate results, document assumptions, and communicate complex analytical findings clearly.

Tata Consultancy Services | Client: BMW

Hyderabad, India

Data Analytics Scientist

Apr 2021 – Dec 2023

- Developed analytics solutions for manufacturing, quality, and operational datasets using Python, SQL, PySpark, Databricks, and Power BI, improving reporting and insight delivery by 40–45%.
- Built predictive models and forecasting workflows to identify equipment/process trends, quality deviations, operational anomalies, and performance improvement opportunities.
- Processed large-scale structured and semi-structured datasets with Spark-based transformations and automated validation checks, reducing manual preparation effort by 35–40%.
- Partnered with engineering and business teams to develop hypotheses, translate technical findings into actionable recommendations, and support data-driven process improvements.

SELECTED PROJECTS

Manufacturing Quality Anomaly Detection | Python, PySpark, scikit-learn, Databricks

- Built a machine learning workflow to prepare process data, engineer quality-related features, and identify anomalous production patterns for earlier investigation.
- Applied model validation and performance monitoring practices, improving anomaly review efficiency by 30%.

Computer Vision Defect Classification Prototype | Python, TensorFlow, CNN

- Developed a CNN-based prototype for image classification and defect-pattern recognition, including preprocessing, training evaluation, and error analysis.
- Documented opportunities for transfer learning and transformer-based vision approaches to improve scale and model accuracy.

LLM-Enabled Research Insight Assistant | Generative AI, RAG, Prompt Engineering

- Designed a RAG-style analytical assistant concept to summarize technical findings from governed project documents and datasets with source-aware outputs.
- Applied prompt engineering, response evaluation, and guardrail practices to improve clarity and reliability of generated insights.

EDUCATION

University of Kansas

Lawrence, KS

Master of Science in Computer Science, GPA: 3.8/4

- **Relevant Coursework:** Machine Learning, Deep Learning, Data Analytics, Statistical Methods, Data Mining, Computer Vision, Big Data Analytics

CERTIFICATIONS

Databricks Certified: Data Engineer Associate | **Microsoft Certified:** Generative AI Developer | **AWS Certified:** Developer Associate